COL1000: Lab 6

Semester 2, 2025–26

Instructions Upload your code to Gradescope as: `<entryNumber>_q<quesNumber>.py`. After uploading, run the autograder to check that all tests pass. Make sure not to include any custom prompt text in `input()` calls, and any extra text in `print()` statements. You must use only basic control structures (loops, recursion, conditionals, functions) and avoid advanced built-in python functions.

Question 1: Count ways to climb stairs (using 1 or 2 steps) Write a program that reads a positive integer `n` and uses a recursive function `ways(n)` to print the number of distinct ways to reach step `n` if a person starts at the ground level (step `0`) and can climb either `1` step or `2` steps at a time. For example, for `n=4`, output must be `5` (different ways are: 1111, 112, 121, 211, 22).

Hint: Obtain an appropriate relation between `ways(n)` and `ways(n-1)`, `ways(n-2)`.

Question 2: Cube-root of Expression Write a program that reads a positive integer `n` and prints $\lfloor (n^2 + 10)^{1/3} \rfloor$ using Binary search. For example, for the input `n=25` the code should print `8`.

Question 3: Prime GCD Write a program that reads a positive integer `n` and computes the number of pairs $1 \leq i < j \leq n$ such that $\gcd(i, j)$ is a prime number.

Question 4: Wildcard pattern matching Write a program that reads two non-empty strings `text` and `pattern`, and uses a recursive function `wildcard_match(text, pattern)` to check whether `pattern` matches the entire `text`.

The pattern may contain:

- `?` which matches **exactly one** character.
- `*` which matches **any sequence** of characters (including the empty sequence).

Your code should print `True` if the match succeeds, and `False` otherwise.

Examples:

- `text = "function"`, `pattern = "f?nct*"` → `True`
- `text = "union"`, `pattern = "*u*o?"` → `True`
- `text = "union"`, `pattern = "u*d"` → `False`
- `text = "abc"`, `pattern = "a??c"` → `False` (needs 4 characters)